

Protective factors against extremism and violent radicalization in youth: A multilevel meta-analysis[☆]

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ABSTRACT

Global concerns about radicalization highlight the need to identify factors protecting youth, which can help in more effective risk assessment and prevention. Most previous research focused on risk factors, but as radicalization is not a linear developmental process, protective factors are highly important. This meta-analysis summarized the overall impact of protective factors against extremism and violent radicalization in youth. In total, 41 studies were included, reporting on 49 independent samples ($N = 55,563$), yielding 194 effect sizes. Potentially protective factors were classified into 12 domains based on similarity. Results of multilevel meta-analyses demonstrated that six domains had a protective impact, including: school achievement ($r = -0.21$), female gender ($r = -0.16$), (pro)social behavior ($r = -0.14$), positive parental involvement ($r = -0.13$), procedural justice ($r = -0.11$), and positive intergroup contact ($r = -0.10$). Moderator analyses revealed differential effects, such that school achievement factors were more protective for right-wing and religious ideologies compared to left-wing ideologies or that positive life outlook factors were more protective in samples with more girls. The identified protective factors are useful for policies to promote positive development in youth to prevent adverse outcomes. More longitudinal research is needed to study direct or buffering protective effects.

[Pre-registration.](#)

1. Introduction

According to the Global Database of the University of Maryland (IEP, 2020), deaths from terrorism have temporarily decreased. However, recent terrorist attacks across the world may have influenced that downward trend (e.g., suicide bombing by Islamic State in Khar, Pakistan; mass shootings by Hamas in Israel, car bombing by Al-Shabaab in Somalia). The Global Peace Index also reports that violent demonstrations have almost doubled since 2008 (IEP, 2022). There are growing concerns about the deterioration of people's trust in social institutions and democratic structures (AIVD, 2023). In part, because this may increase risks for extremism and violent radicalization (Doosje et al., 2016; LaFree, 2023; Lösel et al., 2018). Young people in particular appear to be at risk (Campelo et al., 2018). There are sound systematic reviews and meta-analyses on risk factors for extremism, (violent) radicalization, and terrorism (e.g., Campelo et al., 2018; Emmelkamp

et al., 2020; LaFree & Schwarzenbach, 2021; Siegel et al., 2019; Wolfowicz et al., 2021), however, the main focus of these reviews is on adult samples. A number of case-oriented risk assessment instruments have been developed, but due to various problems their validity seems to be rather limited (e.g., Hassan et al., 2022; RTI International, 2017; Scarcella et al., 2016). For example, there are problems of definition, heterogeneity of phenomena, data access, social desirability, and a (plausible) scarcity of large-scale prospective longitudinal studies. In recent years, it has also become increasingly obvious that the focus on risk factors needs to be complemented by information about protective factors against (violent) radicalization. This multilevel meta-analysis was therefore conducted to integrate current literature and quantify the impact of protective factors on extremism and violent radicalization for adolescents and young adults (i.e., mean age 25-years-old or below).

Various process models of the development of extremism and radicalization assume a more or less linear progress towards an increase of

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seriousness and violence. Popular examples are the Three-N-Concept (Webber & Kruglanski, 2017) or the Staircase Model (Moghaddam, 2005). Similarly, a group of already radicalized individuals may get deradicalized and/or disengaged as described in the Attitudes-Behaviors-Corrective Model (Khalil et al., 2019). These processes are due to the influence of protective factors or “shields” of resilience (Doosje et al., 2016; Lösel et al., 2018).

A protective factor reduces the probability of an adverse outcome (Lösel & Bender, 2003; Lösel & Farrington, 2012). Protective factors are essentially similar to risk factors, but often the positive pole of a continuous risk dimension or the absence of a dichotomous risk. To reduce tautology it is helpful to differentiate between direct protective (promotive) factors and buffering protective factors (Lösel & Farrington, 2012). Direct protective factors have a desirable influence on development without regarding other factors (main effect), whereas buffering protective factors moderate the relation between a risk factor and the behavioral outcome (interaction effect).

There are no generally accepted definitions of extremism and radicalization (e.g., Lösel et al., 2020; Maskaliūnaitė, 2015). *Extremism* can be seen as the verbal or active opposition to basic values in society, such as (in Western countries) democracy, equality, liberty, rule of law, and tolerance for the faiths and beliefs of others. *Radicalization towards violence* is a process by which a person adopts beliefs that justify the use of violence for social and/or political change (Doosje et al., 2016; Friedrich Lösel et al., 2018). *Terrorism* is an intentional act or threat of violence by a non-state actor that aims to attain a political, religious, economic, or social goal and coerces an audience beyond the direct victims (Lösel et al., 2020).

There are also different ideological backgrounds or types of extremism and radicalization (Sarma, 2017; Wolfowicz et al., 2021), for example, right- and left wing political, religious, ethnic, nationalistic, and special issues of motivation (e.g., anti-abortion, ecology preservation). These thematic motivations sometimes overlap (e.g., in ethnic and religious motivation) and there seem to be similar developmental processes by focusing on a shared, perceived threat and designating a hostile outside group responsible for the wrongdoing (Ebner, 2017). There is also some evidence for differences in risk and protective factors depending on the type of ideology, such as political, religious or other ideologies (Wolfowicz et al., 2021).

Although research on risk and protective factors is not primarily theory testing, the investigated factors are embedded in various psychological, social, political, and criminological theories that explain why people may violently radicalize. The Social Strain theory, for example, suggests that because of strain caused by grievance or negative events, particular groups may feel disadvantaged in society. In response to this (perceived) unfair treatment, some groups may consider corrective violence against a perceived wrongdoer as a justifiable measure (Agnew, 2010, 2017). Bandura's (1999) Moral Disengagement Theory proposes that people may violently radicalize when they disengage from their internal moral standards and start justifying violence without self-blame. This can be due to a series of cognitive mechanisms such as the dehumanization of targets (i.e., seeing them as less human) or displacement of responsibility (i.e., making only others responsible for one's actions; Aly et al., 2014; Lösel et al., 1990; Ozer & Bertelsen, 2020; Pauwels & De Waele, 2014). Other theories have also been suggested as useful frameworks to better understand the violent radicalization process, including Self-Control Theory, Social Control Theory, Social Learning Theory, Situational Action Theory, and the General Aggression Model (Becker, 2021; Holt et al., 2018; Lösel et al., 2018; Wolfowicz et al., 2021). Yet, as in other fields of violence research, these are not alternative theories but typically explain specific aspects of the violent outcome.

For the present article, we assume a preliminary consensus in which extremism and violent radicalization result from an interaction and accumulation of risk factors in the absence of protective factors (Farrington et al., 2016; Lösel et al., 2018; Wolfowicz et al., 2021). These

factors can be present at the individual (micro), group (meso), and societal (macro) level (Doosje et al., 2016; LaFree & Schwarzenbach, 2021). Although certainly not applicable to all individuals, the violent radicalization processes often occur in stages or phases (Doosje et al., 2016): for example, becoming sensitive to a radical ideology (1), joining a radical group (2), turning to action, and using violence against other groups (3). At each phase, people may be safeguarded from radical influences by protective factors. As most individuals who endorse extremist views or even violent radical attitudes do not exhibit violent behavior (McCauley & Moskalenko, 2008), it is important for prevention to learn more about the factors that protect youth from violent radicalization (Jugl, Bender, et al., 2021).

A systematic review by Lösel et al. (2018) based on 21 quantitative analyses found replicated evidence for several protective domains: individual (e.g., self-control, adherence to the law, police legitimacy), family (e.g., positive parenting behavior, non-violent significant others), educational (e.g., good school achievement), and social environment (e.g., non-violent peers, contact to foreigners, basic attachment to society). Another meta-analysis (Wolfowicz et al., 2021) demonstrated that protective factors can differ depending on the type of (violent) radicalization outcome, for attitudes (e.g., self-esteem, social support, life satisfaction), intentions (e.g., openness, conscientiousness) and behavior (e.g., law legitimacy, school bonding). Overall, there appears to be an overlap in protective factors for violent radicalization and other forms of violence (Lösel et al., 2020).

Most research on the topic has focused on all age groups with the majority of studies being conducted on adults (Lösel et al., 2018; Wolfowicz et al., 2021). In contrast, some systematic reviews and meta-analyses have zoomed in on risk factors for extremism and violent radicalization in youth (Campelo et al., 2018; Emmelkamp et al., 2020). An emphasis on this younger age group is important because radicalization is most prevalent among adolescents and young adults, with increasingly younger starting ages (Siegel et al., 2019). Adolescents and young adults are thought to be particularly vulnerable for extremism and violent radicalization because of their developing identities: radical groups may offer a sense of identity, belonging, and adventure (Siegel et al., 2019). In addition, they are vulnerable to misinformation in support of the radical narrative due to impaired literacy skills and involvement in the digital world, which is often where recruitment takes place (Breakstone et al., 2021; Duindam et al., 2025; Ebner, 2017; Koehler, 2020; Prievara et al., 2019; Siegel et al., 2019).

Given the above-mentioned conceptual framework and the fast increase of extremism studies, an up-to-date overview of the factors that might protect young people from extremism and violent radicalization is urgently needed. Accordingly, we carried out a multilevel meta-analysis. Our *first goal* is to test the impact of protective factors for extremism and violent radicalization in adolescents and young adults (i.e., mean age 25-years-old or below). By applying a three-level random effects model we aim to include all effect sizes per primary study, thereby ensuring that the most complete picture of the topic is provided under optimal levels of statistical power (Assink et al., 2019). This is particularly important in the field of radicalization, as primary studies often include few (violently) radicalized people, making generalization difficult (Emmelkamp et al., 2020).

Primary studies might differ in terms of study design (i.e., cross-sectional versus longitudinal), sample (i.e., general versus criminal), outcomes (i.e., radical attitudes, intentions, or behaviors), protective factor type (i.e., static versus dynamic), and protective factor domain (e.g., school achievement, parent involvement). Therefore, the *second goal* of the current meta-analysis is to provide a more fine-grained picture of which protective factors are relevant under what circumstances by conducting moderator analyses based on study, sample, and outcome characteristics.

2. Methods

2.1. Selection criteria

For the current meta-analysis, a systematic literature search was conducted to select studies that examined the association between protective factors and extremism or violent radicalization. The studies had to meet the following inclusion criteria: 1) studies had examined protective effects for extremism or violent radicalization, including (violently) radical attitudes, radical willingness, and radical behavior 2) studies had reported on the association between protective factors and extremism or violent radicalization, based on quantitative bivariate/multivariate analyses or provide enough information to allow for the calculation of a bivariate test statistic, 3) studies had at least one protective factor (i.e., factor associated with a decreased likelihood of extremism or violent radicalization) on which a hypothesis was formulated or expected protective effect was described, 4) studies had to include a sample of youth and young adults (i.e., mean age of the participants between 12 and 25 years old), and 5) studies had to be written in English or Dutch (i.e., the languages spoken fluently by the first and second author, to ensure accurate study interpretation).

2.2. Search strategy

Relevant publications were identified using the electronic databases PsycINFO, Web of Science, and Criminal Justice Abstracts. Studies were included until April 2023, covering evidence from the period 1995–2023. We used the following four search categories: radicalization, age, protective factors, and quantitative studies. These search categories were each transformed into their own strings, and the total search per database existed of a combination of the four search strings. In addition, an alternative search was conducted which included a Google Scholar search (i.e., examination of the first 200 hits) in March 2023, to optimize the chance of finding unpublished manuscripts (e.g., reports, conference proceedings, posters, master's theses) and ensure no published manuscripts were missed. Finally, the reference lists of relevant meta-analyses and systematic reviews (Campelo et al., 2018; Emmelkamp et al., 2020; Lösel et al., 2018; Wolfowicz et al., 2021) were examined, in addition to the reference lists of all the included studies, to check whether any relevant additional studies could be identified. For an overview of the search strings and the Google Scholar search terms, see Appendix A.

2.3. Study selection

A two-step approach was taken to screen the studies. The first step included the screening of titles and abstracts (i.e., first screening), which was done separately by the first and second author, the second step involved full-text screening (i.e., second screening).

For the first screening, the tool ASReview was used for screening prioritization. ASReview is an open-access software program (van de Schoot et al., 2021) that allows researchers to efficiently scan a large number of studies based on titles and abstract by presenting the studies that it deems most relevant first through active learning. Five key records were selected (Boehnke et al., 1998; Feddes et al., 2015; Miconi et al., 2020; Pauwels & Svensson, 2017; Sklad et al., 2022), to provide the model in ASReview with background knowledge on relevant studies that match the research topic of the current meta-analysis. This started the active learning cycle that allowed ASReview to assess the relevance of the studies and present suggestions of plausibly relevant studies. Authors were presented with studies through the active learning cycle, and selected the studies that seemed to meet the inclusion criteria based on the title and abstract. Through active learning and based on the authors' choices on the relevance of the studies, the model continuously improved itself, trying to present all relevant manuscripts as early on as possible in the reviewing process.

To determine how many titles and abstracts should be screened, stimulation studies were consulted to choose a data-driven strategy. Even though ASReview is a relatively new tool, elaborate stimulation studies have demonstrated that 95% of eligible studies can be found after screening between 8% to 33% of the studies. In addition, these studies determined that after reading 10% of the abstracts, 70% to 100% of relevant abstract were detected (van de Schoot et al., 2021). For the current meta-analysis, the following data-driven strategy was chosen: the first and second authors independently screened at least 1500 titles and abstracts (10%) and continued until at least 200 articles in a row were deemed "irrelevant".

2.4. Coding of studies

A detailed coding form (see Appendix B) was used to extract information about the study, sample, and outcome characteristics from primary studies in a cohesive and reliable manner. Examples of *study characteristics* included publication year (continuous variable), publication status (published or not published), type of article (e.g., journal article, technical report, dissertation), study design type (e.g., longitudinal, cross-sectional) and country (North America, other). Coded *sample characteristics* included but were not limited to: mean age (continuous), socio-economic status (SES), target group (i.e., community, at-risk, combined), proportion of boys, and proportion of minorities. A selection of assessed *outcome characteristics* included type of (violent) radicalization (i.e., attitudes, intentions, behavior), radical ideology (e.g., political, religious), and type of protective factor (i.e., static, dynamic). For most studies, it was not possible to code all characteristics. For example, studies rarely mentioned the SES of the sample. All studies were coded by the second author (i.e., an assistant-professor with extensive prior coding experience). A subsample of the studies was double coded by the first author, resulting in good inter-rater reliability (IRR = 79% for a subset of 20% of the studies, and complete agreement on factors such as study type, radicalized behavior, target group, sample age, sample gender, publication year). Any coding disagreements were discussed until consensus was reached.

All protective factors were organized into a protective domain. A domain is a group of relatively homogeneous protective factors that can be classified into one domain based on similarity. By distinguishing different protective domains, an overall effect of each protective domain for radicalization could be estimated. The following 12 protective domains were distinguished (see Appendix C): *gender* (being female); *school achievement* (factors related to high education and school achievement); *intergroup contact* (factors related to interethnic friendship and egalitarianism); *parental involvement* (factors related to parental control and bonding); *positive life outlook* (factors related to future orientation and meaning in life); *procedural justice* (factors related to perceived procedural and social justice); *(pro)social behavior* (factors related to empathy and social competence); *religiosity* (factors related to religiosity and spirituality); *positive school environment* (factors related to safety and connectedness at school); *skills* (factors related to self-control and cognitive competence); *social connectedness* (factors related to sense of community membership and inclusion within society); and *well-being* (factors related to satisfaction with life and resilience). Three effect sizes could not be categorized into a domain, and were therefore not analyzed (i.e., time in prison, leftness of teachers, and national identity).

Finally, to assess study quality the Cambridge Quality Checklist was used (Murray et al., 2009, see Appendix B). Studies were rated (range 0–15) based on the quality of several study aspects, such as sampling (i.e., random samples were rated as high quality, whereas convenience sample or case-control were rated low quality), response and attrition rates (i.e., with >70% and <10% respectively being of high quality), design (i.e., longitudinal design receiving the highest quality score), and the reliability coefficient of the assessment tools. A total quality score was calculated based on the mean score of the quality ratings of the different study aspects.

2.5. Calculation of effect sizes

The included studies reported different effect sizes (e.g., beta's, t -values, and correlations). To generate a statistic that summarizes the effects of the protective factor on participants' extremism or violent radicalization, the Pearson's bivariate correlation coefficient (r) was used. If studies did not report correlation coefficients, the corresponding author was emailed to request this information. If authors did not respond or if information could not be provided, correlations (r) were calculated based on other test statistics (i.e., means and SDs), using an online effect size calculator (Wilson, n.d.). Standardized regression coefficients (β) were transformed into correlations (r) by using the following formula: $r = \beta + .05\lambda$ (Peterson & Brown, 2005). The indicator variable of 1 was used for λ when β was positive and 0 when β was negative. If the author(s) did not provide enough information to calculate an effect size but did report a non-significant result, an effect size of zero was assigned, following the procedures of recent multilevel meta-analyses (e.g., Emmelkamp et al., 2020). This occurred for two effect sizes. To facilitate correct interpretation of the findings, a negative sign was given to correlation coefficients when protective factors were associated with lower levels of extremism or violent radicalization, whereas a positive sign was assigned when protective factors were related to higher levels of extremism or violent radicalization outcomes. Since correlations are non-normally distributed (e.g., Lipsey & Wilson, 2001), we transformed the r values into Fisher's z values. After the analyses, we transformed the Fisher's z values back into correlations to make interpretation of the findings easier. A very small effect size was considered $r \geq -0.05$, a small effect size $r \geq -0.10$, a medium effect size $r \geq -0.20$, and a large effect size $r \geq -0.30$ (Funder & Ozer, 2019).

2.6. A three-level random effects meta-analytic model

For the current meta-analysis, a three-level random-effects model was chosen because it can account for within-and between study variation, which is useful when different studies, each with their unique methodologies and samples, are included in a meta-analysis (Van Den Noortgate & Onghena, 2003). By modeling the different levels of variance, multiple effect sizes from the same study can be included, which is preferred over hand-picking or averaging, as this results in the loss of information and lower statistical power (Assink et al., 2019; Assink & Wibbelink, 2016; Lipsey & Wilson, 2001). The three-level random effects model accounts for sampling variance of effect sizes (level 1), variance between effect sizes from the same study (level 2), and the variance between studies (level 3; Assink & Wibbelink, 2016). As effect sizes are considered a random sample of the population of effect sizes in a random-effects model, findings can be generalized to the larger population of interest (Kelley & Kelley, 2012).

2.7. Overall effect

Analyses were conducted in Rstudio (RStudio Team, 2023), including the following packages: foreign, metafor, influence.ME, lme4, and Matrix. As a first step, the overall effect of protective factors on extremism and violent radicalization was computed through an intercept only-model, using procedures described by Assink and Wibbelink (2016). Study results were weighted based on sample size; no corrections were made to the raw data (Hunter & Schmidt, 2004). All protective factors, regardless of their domain, and all outcomes' variables of extremism and violent radicalization were combined to assess the overall effect of protective factors.

2.8. Protective factor domains: overall effects and moderator analyses

Next, overall effects for each protective factor domain separately were computed through an intercept only-model, resulting in an independent overall effect for each of the 12 domains. Using procedures

described by Assink and Wibbelink (2016), significant heterogeneity was assessed. Log-likelihood tests were performed to examine to what extent there was significant heterogeneity between effect sizes from the same studies (level 2) and between studies (level 3). If there was significant heterogeneity for at least one of the levels, moderator analyses were performed by including the potential moderators in each protective factor domain model (Assink & Wibbelink, 2016).

Data were prepared for moderator analyses by centering continuous (moderator) variables around their mean. In addition, dummy variables were created for the categorical variables. Tested moderators included study characteristics (publication year, impact factor, study quality, country, design type), sample characteristics (mean age, proportion of minorities, proportion of boys), and outcome characteristics (radicalization type, violent radicalization, radical ideology, type of protective factor). Finally, for some protective factor domains (e.g., procedural justice, social connectedness), specific moderators were tested (see Appendix C).

2.9. Sensitivity analyses

Finally, sensitivity analyses were conducted to assess the influence of individual studies' on the overall effect of protective factors found. To examine the influence of one individual study on the overall effect of protective factors on extremism and violent radicalization, robustness tests were conducted by recomputing the overall effect after excluding effect sizes from one study at a time (Viechtbauer & Cheung, 2010). Moreover, for two effect sizes (of two studies) a zero was assigned, when authors had reported a non-significant result (and not provided enough information to calculate an effect size). To assess the impact of this adjustment, the overall effect was recalculated excluding these effect sizes. Finally, we recalculated the overall effect by winsorizing extreme effect sizes (effect sizes 1.5 times the interquartile range), that is replacing the outlier by the lowest or highest score within the normal range (e.g., Emmelkamp et al., 2020).

2.10. File drawer problem

To assess to what extent the findings of this meta-analysis are influenced by the file drawer problem (i.e., non-significant findings being less likely to get published; Rosenthal, 1979), we performed the Rosenthal's fail-safe test. When the fail-safe N exceeds the critical value (of $5 \times k + 10$, where k is the number of independent samples; in this study the critical value is 255), there is no indication of the file drawer problem or publication bias (Rosenthal, 1979). Moreover, we analyzed to what extent effect sizes were normally distributed by computing a funnel plot. An asymmetrical funnel plot is indicative of the file drawer problem or publication bias. To measure the degree of asymmetry in the funnel plot, we performed an Egger's regression test for multilevel data, by assessing the likelihood of a significant relationship between effect size and standard error (e.g., Abdulla Alabbasi et al., 2023). Finally, we also performed a trim and fill procedure (Duval & Tweedie, 2000a, 2000b). Previous studies showed that effect size estimates based on imputation of effect sizes after this procedure might not be accurate for multilevel meta-analyses (Fernández-Castilla et al., 2020). Hence, we used the procedure of Fernández-Castilla et al. (2020), which estimates the effect sizes imputed at the left or right side of the distribution. It is important to consider the effect of potential missing data, as it is unlikely in a meta-analysis that all relevant studies are retrieved; this can positively – or negatively – bias findings (Duval & Tweedie, 2000a, 2000b).

3. Results

3.1. Study selection

In total, the search yielded 15,749 studies after the removal of duplicates (see Fig. 1). After screening the title and abstracts (i.e., first

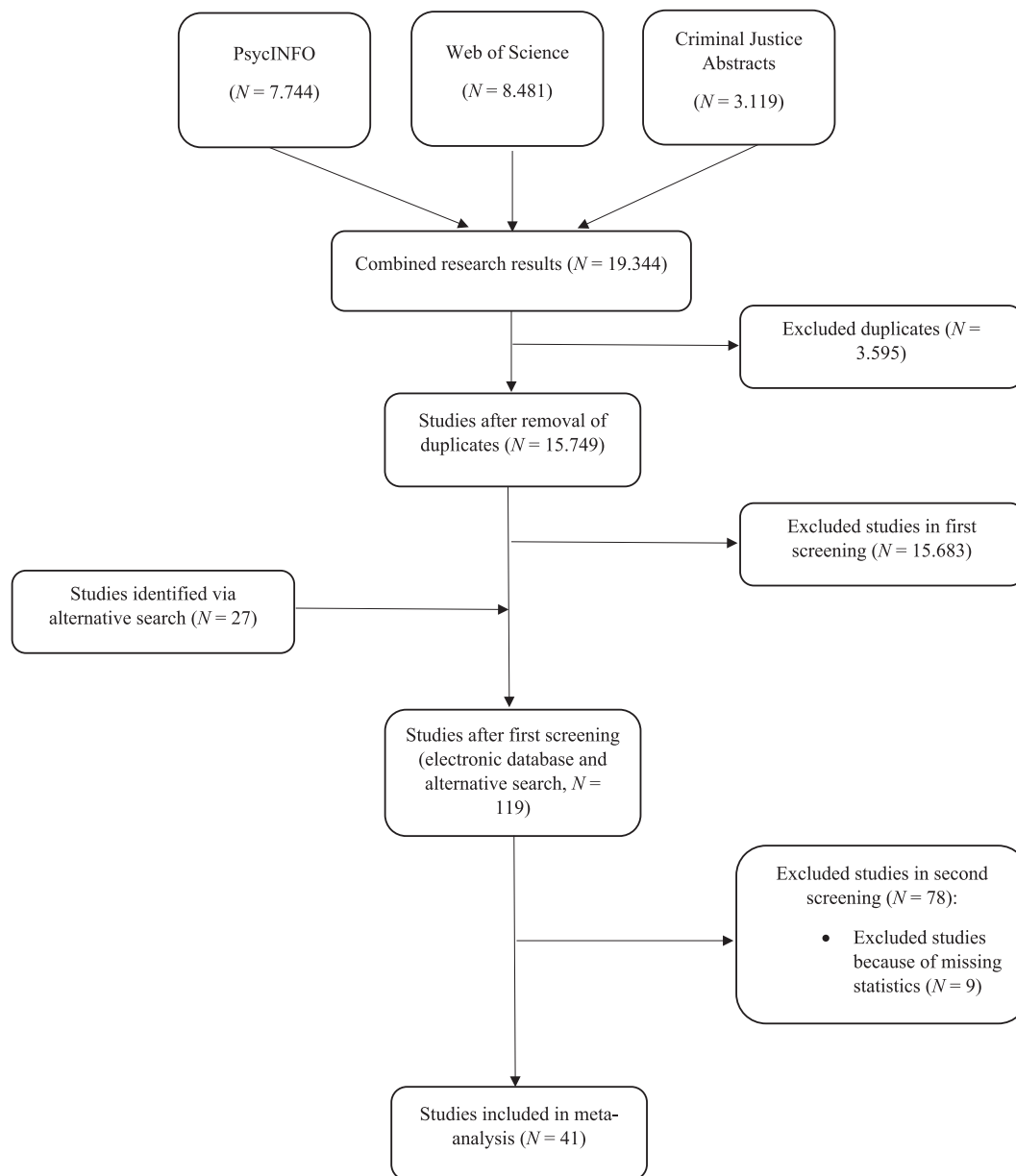


Fig. 1. Flow chart.

screening) in ASReview and conducting the alternative literature search, 119 studies were identified as relevant for full text screening. After the second full text screening, 78 publications were excluded. Nine of these 78 publications had missing test statistics. Several attempts were made to contact the corresponding author to retrieve test statistics, with no success for these publications. Overall, this resulted in a final number of 41 included publications reporting on $k = 49$ independent samples. The included studies produced 194 effect sizes, each reflecting the effect of a protective factor on extremism or violent radicalization. Most protective factors in the current meta-analysis were putative, meaning that they did not precede extremism or violent radicalization. The total sample included $N = 55.563$ juveniles. Table 1 presents an overview of the included studies and their characteristics. The included studies are marked with an asterisk (*) in the reference list.

All the studies included in this multilevel meta-analysis were issued between 1995 and 2023 and all were peer reviewed. Across samples, the average age was 18.74 years ($SD = 3.29$; based on 33 samples). Approximately half of the sample was male (49.4%; based on 44 samples) and had a minority background (52.8%; based on 25 samples).

The average impact factor of the journals of the included studies was 3.3 ($SD = 1.8$; based on 48 samples). Study quality ranged from 2 to 11, with a mean of 4.02 ($SD = 1.52$). Most studies were conducted outside of North-America ($N = 32$ studies, 38 samples, 154 ES), were from Western countries ($N = 36$ studies, 43 samples, 157 ES), were cross-sectional ($N = 37$ studies, 45 samples, 181 ES), consisted of dynamic protective factors ($N = 38$ studies, 45 samples 189 ES), and consisted of violent outcomes (instead of nonviolent outcomes; $N = 32$ studies, 39 samples, 173 ES). About half of the studies included attitudinal outcomes ($N = 21$ studies, 23 samples, 68 ES), whereas the other half reported intention or behavioral outcomes ($N = 20$ studies, 26 samples, 126 ES). Regarding ideology, five studies focused on right-wing ideology (6 samples, 23 ES), seven studies on religious ideology (8 samples, 37 ES), five studies on group or organizations (7 samples, 23 ES), 17 studies on political ideology (21 samples, 90 ES), and nine studies were unspecified (9 samples, 32 ES).

Table 1
Detailed description of the included studies.

Authors	N	Mean age (SD), age range	% of males	% of minorities	Radicalization outcome(s)	Protective factor(s)	No. of ES	Quality (0–15)
Van Bergen et al., 2015	1370	15.86 (1.02), 14–18	47	100	Attitude to use violence as means to defend the in-group (ATT, REL), willingness to use violence as means to defend the in-group (INT, REL)	Psychological connectedness to Dutch society (SOC), out-group friendship (SCH)	8 P	2–3
van Bergen et al., 2016	970	15.58 (0.95), 14–18	48	100	Attitude towards violent in-group defense by others (ATT, REL), willingness to use violence in defense of the in-group (ATT, REL)	Egalitarian socialization (FAM), connectedness with mainstream society (SOC)	4 P	2–3
Boehnke et al., 1998	590	Seventh and ninth graders	Close to equal proportions of girls and boys	N/A	Right-wing extremism (ATT, RW)	Parental control (FAM), academic success (SCH)	2 P	7
Bovier & Boehnke, 1999	694	15.42 (0.68)	N/A	N/A	Right-wing attitudes (ATT, RW)	Leftness of teachers (SCH), (permissively) positive social climate in the classroom (SCH)	2 NP	6
Cardeli et al., 2020	532	22.15 (3.09), 18–35	61	100	Attitudes towards violent and/or illegal political activism (ATT, POL)	Neighborhood intergeneration closure (SOC), neighborhood social cohesion (SOC), neighborhood informal social control (SOC), social connectedness to online community (SOC)	4 NP	3–4
Costabile et al., 2020	328	16.67 (1.50)	55	3	Radicalism (INT, UNSP)	Psychological wellbeing (IND)	1 P	3
Del Pino-Brunet et al., 2023	213	18.26 (2.67), 15–25	100; 0	15	Keeness to participate in legal, nonviolent, illegal or violent political actions and individual's tendency towards self-sacrifice (INT, POL)	Satisfaction with life (IND)	2 P	3
Eldor et al., 2022	334	N/A, 16+	32.2	26.2	Willingness to participate in radical activities in society (INT, UNSP), violent intentions (INT, UNSP)	School attentiveness (SCH), equality (SCH), mutual respect (SCH), safety and connectedness (SCH)	8 P	4
Ellis et al., 2015	79	20.76 (2.16), 18–25	100	100	Attitudes towards illegal and violent activism (ATT, POL)	Perceived importance of political and social groups (family (FAM), Somalia (SOC), religious group (REL), clan/tribe (SOC), US (SOC), psychological sense of community membership (acceptance and belonging US and Somali; SOC), posttraumatic growth (IND)	2 P, 8 NP	3–4
Ellis et al., 2016	374	21.3 (2.90), 18–30	62.3	100	Readiness to participate in illegal and violent political action (INT, POL)	Perceived importance of social groups (tribe; SOC), psychological sense of community (Somali and American/Canadian; SOC), attachment to US/Canada (SOC)	4 NP	4–5
Ellis et al., 2021	593; 95; 103	N/A, 18+ (85.4% between 19 and 24); N/A, 18+ (69.5% between 19 and 24); N/A, 18+ (65% between 19 and 24)	30.6; 63.2; 54.4	37.6; 100; 100	Readiness to participate in illegal and violent behavior for one's group or organization (INT, GR)	Perceived social support (SCH & FAM)	3 P	4; 3; 3
Feddes et al., 2015	46	16.93 (2.76), 14–23	73	100	Attitude towards ideology-based violence by others (ATT, REL), own violent intentions (INT, REL)	Empathy (IND), perspective taking (IND)	4 P	3–4
Hagan et al., 1995	489	N/A, 7–9th graders	N/A	N/A	Right-wing orientations (ATT, RW)	Parental control (FAM), school achievement (SCH)	2 P	8
Hagan et al., 1999	545; 1684	14.53 (1.15), 7–9th graders; 14.73 (1.21), 7–9th graders	50; 41	N/A	Affiliations with right-wing extremism (ATT, RW)	Female gender (UNSP)	2 P	4; 4
Jahnke et al., 2020	303; 6715	24.23 (3.32), 18–30; 14.60 (0.65), 13–18	47.9; 48	N/A; 0	Readiness to participate in illegal or violent acts (INT, POL); general radicalization (ATT, UNSP), attitudes towards the use of violence among right-wing groups (ATT, RW)	Observer sensitivity (IND)	3 P	2; 4
Jahnke et al., 2022	6715	9th graders	47	44	Willingness to engage in political violence (INT, POL)	Female gender (UNSP)	1 P	4
Kaczowski et al., 2022	340	24.32 (1.90), 18–29	100	16.8	Violent extremist intentions (INT, POL), attitudinal support	Social network diversity (SOC)	10 P	3

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Table 1 (continued)

Authors	N	Mean age (SD), age range	% of males	% of minorities	Radicalization outcome(s)	Protective factor(s)	No. of ES	Quality (0–15)
Kamenowski et al., 2021	8317	N/A, <18 (55.8% are 17 or 18)	49.7	52	for violent extremism (ATT, POL) Left-wing extremism (ATT & INT, UNSP), Islamist extremism (ATT & INT, REL), right-wing extremism (ATT & INT, RW) Political violence (BEH, POL)	Christian religiosity (REL), high education (SCH), high parental control (FAM), high moral values (IND), religious tolerance (REL)	18 NP, 8 P	4–5
Khoury-Kassabri et al., 2015	167	16.84 (1.77), 15–21	100	N/A	Religious militancy (ATT & INT, REL)	Parental control (attachment and involvement; FAM), Religiosity: relationship, request, retribution, ritualism (REG)	5 P	3–4
King et al., 2021	342	19.82 (2.46), 15–27	100	N/A	Radicalism (INT, GR)	Religious socialization (REL), current religiousness (REL), time in prison (UNSP)	3 P	3–4
Koirikivi et al., 2021	2837	17.20 (N/A), 15–19	41.6	N/A	Readiness to use violence in political action (INT, POL)	Resilience: self-regulation skills (IND), peer relations (SCH), caregiver support (FAM), school safety and connectedness (SCH), meaning in life (IND), satisfaction with life (IND), empathy (IND), beneficence (IND)	8 P	4–5
Kuhn, 2004	1309	18+ (40% 18, 59% 19, 1% 20+)	38.0	N/A	Support for terrorist organizations (ATT, REL), support for the World Trade Center attack (ATT, UNSP)	Communication with parents about politics (FAM), positive attitude (IND), future plans (IND)	3 P	4–5
Levin et al., 2003	360	20.56 (2.56), 18–34	50	N/A	Sympathy for violent radicalization (ATT, UNSP), radical intentions (INT, GR)	Social dominance orientation (SOC)	2 P	4–5
Miconi et al., 2020	1680	N/A, 16+ (74% between 16 and 21)	29	27	Sympathy for violent radicalization (ATT, UNSP), radical intentions (INT, UNSP)	Future orientation (IND)	2 NP	3–4
Miconi et al., 2021	931	N/A, 16+ (81% between 16 and 24)	33	34	Support for violent radicalization (ATT, GR)	Intrinsic religiosity (REL)	2 NP	4–5
Miconi et al., 2022	3100	18.60 (1.76), 16–25	27.5	30.6	Willingness to participate in high-risk political action likely to violate laws and social standards (INT, POL)	Future orientation (SCH), presence of meaning (SCH), search for meaning (SCH)	3 P	5
Moreira et al., 2018	251; 201	23.06 (3.76), 18–30; 19.52 (1.81), 18–27	40.6; 26.4	N/A	Support for political violence (ATT, POL)	Perception of social justice (IND)	2 NP	4; 4
Moyano et al., 2022	120	N/A	N/A	N/A	Generic support for violent extremism (ATT, UNSP)	Social support (SOC)	4 P	3–5
Nivette et al., 2017	1214	15.0 (N/A), N/A	N/A	76.2	Extremism (ATT, UNSP), acceptance of the use of violence in relation to extremism (ATT, UNSP), acceptance of the use of illegal means in relation to extremism (ATT, UNSP)	Generalized trust (SOC), parental involvement (FAM), conflict coping skills (IND), other religious background (non-Muslim) or non-religious background (REL)	4 P	11
Ozer, 2020	223	22.74 (2.32), 18–29	28.3	N/A	Acts of violence towards property for political or religious reasons (BEH, REL & POL), acts of violence towards persons for political or religious reasons (BEH, REL & POL), support for right-wing extremism for political goals (ATT, RW)	Multicultural acquisition (IND)	3 P	4
Pauwels & De Waele, 2014	2879	N/A, <22	35.9	N/A	Political violence: acts of violence committed towards persons and property because of political or religious beliefs (BEH, REL & POL), right-wing extremism (ATT, RW)	Social integration (SOC), perceived procedural justice (IND), perceived legitimacy (IND)	9 P	4–5
Pauwels & Svensson, 2017	6020	20.21 (N/A), 16–24	N/A	N/A	Willingness to commit property damage on behalf of a militant animal protection organization (INT, GR & POL), willingness to commit personal damage on behalf of a militant animal protection organization (INT, GR & POL), radicalism intention	Ability to exercise self-control (IND)	2 NP	3
Pfundmair et al., 2021	110	16.46 (N/A), grade 9, 11, and 12	33.6	9.1		Internalized moral identity (IND), symbolized moral identity (IND)	6 P	3–4

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Table 1 (continued)

Authors	N	Mean age (SD), age range	% of males	% of minorities	Radicalization outcome(s)	Protective factor(s)	No. of ES	Quality (0–15)
Rousseau et al., 2019	1742	N/A, 16+ (83% between 16 and 24)	30	35	pertaining to illegal and violent behavior (INT, GR) Willingness to support illegal and violent behavior in the name of one's group or organization (INT, GR)	Religiosity (REL), perceived social support (SCH & FAM)	2 NP	5
Schmid, 2012	839	Around 16	100; 0	N/A	Readiness for political violence (INT, POL)	Social responsibility (SOC)	2 P	3
Schröder et al., 2022	6715	14.60 (0.70), 13–19	47.1	41.9	Right-wing attitudes (ATT, RW)	Family cohesion (FAM), parental control (FAM)	2 NP	3
Shechory & Laufer, 2008	163	16.20 (1.18); N/A	39,3	N/A	Illegal ideology activity (blocking main traffic arteries, arrests, clashes with military/police forces; BEH, POL)	Attachment (FAM), involvement (FAM), commitment (IND), belief (SOC)	4 P	2–3
Simon et al., 2013	845	24.0 (3.75), 18–46	33	100	Sympathy for radical action (ATT & INT, UNSP), radical activity in the past (BEH, UNSP)	Duel identification (IND)	3 P	3–6
Tausch et al., 2011	332; 157	22.79 (3.37), N/A; 21.66 (2.06), N/A	44.0; 47.1	N/A	Likelihood to participate in actions against tuition fees in the future (INT, POL); support for ingroup violence (ATT, REL)	Perceived group efficacy (SOC)	2 P	4; 3
Vergani et al., 2019	148	21.6 (3.51), N/A	37.2	N/A	Militant extremism (ATT, UNSP)	Female gender (UNSP)	1 NP	3
Yu et al., 2023	2016	13.92 (1.10), 12–17	49.5	0.49	Illegal violent political action (BEH, POL)	National identity (UNSP), self-determination (IND), self-efficacy (IND), clear and positive identity (IND), beliefs in the future (IND), prosocial involvement (SOC), prosocial norms (IND), social responsibility (IND), spirituality (IND), parental behavioral control (FAM), bonding (SOC), positive/high parent-child relationship (FAM), family mutuality (FAM), family communication (FAM), opportunities to discuss socio-political issues at home (FAM), opportunities to discuss socio-political issues at school (SCH), school's collective views against violence (SOC), resilience (IND), social competence (IND), recognition of positive behavior (IND), emotional competence (IND), cognitive competence (IND), behavioral competence (IND), moral competence (IND)	25 P	5–6

Radicalization outcomes: ATT = attitudes, INT = intentions, BEH = behavior; ideologies: RW = right-wing, REL = religious or ethnic, GR = group or organization, POL = political; UNSP = unspecified; protective factor domain: SCH = school and friends, FAM = family, IND = individual/cognitive, SOC = societal, REL = religious or ethnic; no. of ES = number of effect sizes, N/A = not reported, P = published, NP = not published.

3.2. Overall effect

Test statistics for the overall effect are presented in Table 2. Results indicated that the three-level model had a significantly better fit over the two-level model omitting between-study variance (LRT = 11.92, $p < .001$), with a higher loglikelihood value (65.71 versus 59.75), and reductions in AIC ($\Delta AIC = -9.92$) and BIC ($\Delta BIC = -6.65$).

The overall effect size of protective factors on extremism and violent radicalization was small and negative ($r = -0.098$, 95% CI -0.130 to -0.066 , $p < .001$), with 87.09% of the variance attributed to within study variance and 10.29% of the variance to between study variance. This overall effect indicates that protective factors marginally decreased extremism and violent radicalization. The heterogeneity test demonstrated that there was significant within- and between-study variance ($p < .001$).

3.3. Overall effects for different protective factor domains

Test statistics for the overall effects per protective domain are presented in Table 3. Six domains of protective factors yielded a significant result (ordering effect sizes from largest to smallest; reporting % variance at level 2 and level 3): school achievement ($r = -0.205$, $p < .001$; 79.9% var. L2, 2.1% var. L3), female gender ($r = -0.156$, $p < .05$; 43.3% var. L2, 43.3% var. L3), (pro)social behavior ($r = -0.139$, $p < .01$; 60.4% var. L2, 37.4% var. L3), positive parental involvement ($r = -0.134$, $p < .01$; 92.9% var. L2, 5.8% var. L3), procedural justice ($r = -0.111$, $p < .05$; 42.4% var. L2, 54.6% var. L3), and positive intergroup contact ($r = -0.103$, $p < .05$; 0.9% var. L2, 59.8% var. L3). The other protective factor domains (i.e., positive life outlook, procedural justice, religiosity, positive school environment, skills, social connectedness, and well-being) were not significantly associated with extremism and violent radicalization.

Table 2

Overall mean effect sizes.

Outcome	N studies (samples)	N ES	Mean z (SE)	95% CI	t-value (mean z)	Mean r
Extremism and violent radicalization	41 (49)	194	-0.098 (0.016)	-0.130—-0.066	-6.059***	-0.098
- Excluding Van Bergen et al. (2015)	40 (48)	186	-0.096 (0.017)	-0.129—-0.063	-6.066***	-0.096
- Excluding van Bergen et al. (2016)	40 (48)	190	-0.095 (0.016)	-0.127—-0.062	-5.802***	-0.095
- Excluding Boehnke et al. (1998)	40 (48)	192	-0.098 (0.016)	-0.130—-0.065)	-5.975***	-0.098
- Excluding Bovier and Boehnke (1999)	40 (48)	192	-0.094 (0.016)	-0.126—-0.063	-5.854***	-0.094
- Excluding Cardeli et al. (2020)	40 (48)	190	-0.102 (0.016)	-0.134—-0.071	-6.371***	-0.102
- Excluding Costabile et al. (2020)	40 (48)	193	-0.098 (0.016)	-0.130—-0.066	-6.027***	-0.098
- Excluding Del Pino-Brunet et al. (2023)	40 (47)	192	-0.099 (0.016)	-0.131—-0.067	-6.076***	-0.099
- Excluding Eldor et al. (2022)	40 (48)	186	-0.097 (0.017)	-0.130—-0.064	-5.808***	-0.097
- Excluding Ellis et al. (2015)	40 (48)	184	-0.100 (0.017)	-0.133—-0.067	-6.000***	-0.100
- Excluding Ellis et al. (2016)	40 (48)	190	-0.100 (0.016)	-0.132—-0.068	-6.127***	-0.100
- Excluding Ellis et al. (2021)	40 (46)	191	-0.099 (0.016)	-0.132—-0.067	-6.098***	-0.099
- Excluding Feddes et al. (2015)	40 (48)	190	-0.097 (0.016)	-0.129—-0.065	-5.923***	-0.097
- Excluding Hagan et al. (1995)	40 (48)	192	-0.097 (0.016)	-0.129—-0.065	-5.936***	-0.097
- Excluding Hagan et al. (1999)	40 (47)	192	-0.097 (0.016)	-0.129—-0.065	-5.924***	-0.097
- Excluding Jahnke et al. (2020)	40 (47)	191	-0.100 (0.016)	-0.133—-0.068	-6.157***	-0.100
- Excluding Jahnke et al. (2022)	40 (48)	193	-0.097 (0.016)	-0.129—-0.065	-5.956***	-0.097
- Excluding Kaczkowski et al. (2022)	40 (48)	184	-0.098 (0.017)	-0.132—-0.065	-5.836***	-0.098
- Excluding Kamenowski et al. (2021)	40 (48)	168	-0.085 (0.015)	-0.115—-0.056	-5.665***	-0.085
- Excluding Khoury-Kassabri et al. (2015)	40 (48)	189	-0.096 (0.017)	-0.129—-0.063	-5.820***	-0.096
- Excluding King et al. (2021)	40 (48)	191	-0.099 (0.016)	-0.131—-0.067	-6.052***	-0.099
- Excluding Koirikivi et al. (2021)	40 (48)	186	-0.098 (0.017)	-0.131—-0.065	-5.824***	-0.098
- Excluding Kuhn (2004)	40 (48)	191	-0.098 (0.016)	-0.131—-0.066	-5.973***	-0.098
- Excluding Levin et al. (2003)	40 (48)	192	-0.095 (0.016)	-0.127—-0.063	-5.886***	-0.095
- Excluding Micconi et al. (2020)	40 (48)	192	-0.097 (0.016)	-0.129—-0.065	-5.929***	-0.097
- Excluding Micconi et al. (2021)	40 (48)	192	-0.099 (0.016)	-0.131—-0.066	-6.026***	-0.099
- Excluding Micconi et al. (2022)	40 (48)	191	-0.098 (0.016)	-0.130—-0.066	-5.965***	-0.098
- Excluding Moreira et al. (2018)	40 (47)	192	-0.096 (0.016)	-0.128—-0.064	-5.910***	-0.096
- Excluding Moyano et al. (2022)	40 (48)	190	-0.101 (0.016)	-0.133—-0.069	-6.192***	-0.101
- Excluding Nivette et al. (2017)	40 (48)	190	-0.096 (0.017)	-0.129—-0.064	-5.833***	-0.096
- Excluding Ozer (2020)	40 (48)	191	-0.097 (0.016)	-0.130—-0.065	-5.916***	-0.097
- Excluding Pauwels and De Waele (2014)	40 (48)	185	-0.095 (0.017)	-0.128—-0.062	-5.663***	-0.095
- Excluding Pauwels and Svensson (2017)	40 (48)	192	-0.095 (0.016)	-0.127—-0.063	-5.855***	-0.095
- Excluding Pfundmair et al. (2021)	40 (48)	188	-0.102 (0.016)	-0.134—-0.070	-6.287***	-0.102
- Excluding Rousseau et al. (2019)	40 (48)	192	-0.098 (0.016)	-0.130—-0.066	-5.990***	-0.098
- Excluding Schmid (2012)	40 (47)	192	-0.095 (0.016)	-0.127—-0.063	-5.854***	-0.095
- Excluding Schröder et al. (2022)	40 (48)	192	-0.098 (0.016)	-0.130—-0.065	-5.957***	-0.098
- Excluding Shechory and Laufer (2008)	40 (48)	190	-0.098 (0.016)	-0.131—-0.066	-5.944***	-0.098
- Excluding Simon et al. (2013)	40 (48)	191	-0.100 (0.016)	-0.132—-0.068	-6.114***	-0.100
- Excluding Tausch et al. (2011)	40 (47)	192	-0.098 (0.016)	-0.130—-0.065	-5.966***	-0.098
- Excluding Vergani et al. (2019)	40 (48)	193	-0.098 (0.016)	-0.130—-0.066	-6.015***	-0.098
- Excluding Yu et al. (2023)	40 (48)	170	-0.105 (0.014)	-0.133—-0.076	-7.212***	-0.105

Table 3

Results for the overall mean effect size for the different protective factor domains.

Protective factor	N studies (samples)	N ES	Mean z (SE)	95% CI	t-value (mean z)	Mean r	LRT (level 2/3)	% var. (level 1/2/3)
School achievement	3 (3)	9	-0.208 (0.019)	-0.253—-0.163	-10.982**	-0.205	21.08***/0.00	18.0/79.9/2.1
Gender (female)	3 (4)	4	-0.157 (0.032)	-0.259—-0.056	-4.941*	-0.156	0.00/0.00	13.3/43.3/43.3
(Pro)social behavior	6 (7)	33	-0.140 (0.049)	-0.233—-0.039	-2.844**	-0.139	775.50***/7.06**	2.2/60.4/37.4
Parental involvement	11 (11)	23	-0.135 (0.044)	-0.226—-0.044	-3.071**	-0.134	1608.20***/0.14	1.2/92.9/5.8
Procedural justice	4 (6)	12	-0.111 (0.050)	-0.221—-0.001	-2.221*	-0.111	90.84***/0.83	3.0/42.4/54.6
Intergroup contact	6 (6)	15	-0.103 (0.036)	-0.181—-0.025	-2.845*	-0.103	0.00/3.04	39.3/0.9/59.8
Skills	5 (6)	10	-0.087 (0.073)	-0.252-0.077	-1.200	-0.087	78.14***/5.99*	1.2/25.6/73.1
Social connectedness	8 (8)	39	-0.083 (0.044)	-0.172-0.007	-1.875§	-0.083	21.93***/8.23**	10.3/25.9/63.8
Positive life outlook	6 (6)	10	-0.044 (0.058)	-0.176-0.089	-0.746	-0.044	175.69***/1.15	1.8/53.6/44.7
Positive school environment	4 (4)	12	-0.023 (0.116)	-0.278-0.232	-0.200	-0.023	1049.44***/0.25	1.3/79.8/18.9
Well-being	8 (11)	12	-0.017 (0.038)	-0.100-0.065	-0.460	-0.017	0.33/1.19	7.8/3.5/88.7
Religiosity	7 (7)	13	-0.005 (0.055)	-0.124-0.114	-0.087	-0.005	0.00/5.88*	3.5/0.0/96.5

Note. N studies (samples) = number of studies and independent samples; N ES = number of effect sizes; mean z = mean effect size Fisher z; SE = standard error; CI = confidence interval, t-value = difference in mean z with zero; LRT = likelihood-ratio test for level 2 and level 3; % var. = percentage of variance explained.

§ $p < .10$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

3.4. Moderator analyses for the different protective factor domains

The heterogeneity tests demonstrated that there was significant within- and between-study variance ($p < .001$) for the following protective factor domains: school achievement, parental involvement,

positive life outlook, procedural justice, (pro)social behavior, religiosity, positive school environment, skills, and social connectedness (see Table 3). Significant results are presented here and summarized in Table 4. Tables with non-significant findings can be found in Appendix D.

Table 4

Results for the moderator analyses on (violent) radicalization and extremism for the different protective factor domains (only significant results).

Moderator	<i>N</i> samples	<i>N</i> ES	<i>B</i> ₀ /mean <i>z</i> (95% CI)	<i>t</i> ₀	Mean <i>r</i>	<i>B</i> ₁ (95% CI)	<i>t</i> ₁	<i>F</i> (<i>df</i> ₁ , <i>df</i> ₂)	<i>p</i>
Study characteristics									
Positive life outlook									
Background								<i>F</i> (1, 8) = 7.729	0.024
Western (RC)	4	7	−0.102 (−0.201—0.003)	−2.371*	−0.102				
Non-Western	2	3	0.125 (−0.035–0.285)	1.801	0.124	0.227 (−0.039–0.415)	2.780*		
Skills									
Background								<i>F</i> (1, 8) = 5.924	0.041
Western (RC)	4	5	−0.171 (−0.320—0.022)	−2.643*	−0.169				
Non-Western	2	5	0.090 (−0.107–0.288)	1.056	0.090	0.261 (0.014–0.508)	2.434*		
Religiosity									
Publication year (continuous)	7	13	−0.037 (−0.110–0.037)	−1.099	–	0.045 (0.017–0.072)	3.582**	<i>F</i> (1,11) = 12.829	0.004
Positive school environment									
Quality assessment	4	12	−0.644 (−1.610–0.323)	−1.484	–	0.987 (0.716–1.259)	8.101***	<i>F</i> (1,10) = 65.629	< 0.001
Social connectedness									
Impact factor (continuous)	8	39	−0.070 (−0.136—0.005)	−2.196*	–	0.051 (0.010–0.091)	2.523	<i>F</i> (1, 37) = 6.364	0.016
Sample characteristics									
Positive life outlook									
% boys (continuous)	6	10	−0.038 (−0.127–0.051)	−0.991	–	1.176 (0.142–2.210)	2.622*	<i>F</i> (1, 8) = 6.876	0.031
% minorities (continuous)	3	7	−0.017 (−0.124–0.090)	−0.410	–	1.549 (0.367–2.732)	3.369*	<i>F</i> (1, 5) = 11.347	0.020
Social connectedness									
Mean age (continuous)	6	32	−0.064 (−0.150–0.023)	−1.501	–	0.032 (0.005–0.059)	2.401*	<i>F</i> (1,30) = 5.766	0.023
Outcome characteristics									
School achievement									
Ideology								<i>F</i> (2, 5) = 15.151	0.008
Right-wing (RC)	2	4	−0.236 (−0.268—0.205)	−19.381***	−0.232				
Religious	1	2	−0.238 (−0.273—0.204)	−17.947***	−0.234	−0.002 (−0.049–0.044)	−0.121		
Other	1	2	−0.149 (−0.183—0.115)	−11.215***	−0.148	0.087 (0.041–0.134)	4.840**		
Social connectedness									
Ideology								<i>F</i> (2, 36) = 5.389	0.009
Right-wing (RC)	1	1	−0.043 (−0.222–0.136)	−0.488	−0.043				
Religious	2	6	−0.278 (−0.416—0.139)	−4.070***	−0.271	−0.234 (−0.461—0.008)	−2.112*		
Other	6	32	−0.032 (−0.095–0.032)	−1.014	−0.032	0.012 (−0.166–0.189)	0.132		
Subdomains									
Social connectedness									
Society/country vs. community/ groups								<i>F</i> (1, 37) = 19.171	< 0.001
Society/country (macro) (RC)	4	19	−0.189 (−0.301—0.077)	−3.406**	−0.187				
Community/groups (meso)	4	20	0.034 (−0.082–0.150)	0.596	0.034	0.223 (0.120–0.326)	4.378***		

Note. *N* samples = number of independent samples; *N* ES = number of effect sizes; *B*₀ = mean effect size Fisher's *z*; CI = confidence interval, *B*₁ = estimated regression coefficient; *t*-values = difference in mean *z* with zero; *F*-value = omnibus test of regression coefficients; *p* = *p*-value of omnibus test; RC = reference category.

* *p* < .05.** *p* < .01.*** *p* < .001.

3.5. Study characteristics

Regarding the domains positive life outlook and skills were only protective in studies conducted in Western countries (*r* = −0.102 and *r* = −0.169, respectively), not in non-Western countries (*r* = 0.124 and *r* = 0.090, *p* < .05, respectively). For religiosity, larger protective effects

were observed for older studies (*z* = −0.037, *p* < .01). For positive school environment, larger protective effects were found for lower quality studies (*z* = −0.644, *p* < .001). Regarding the domain social connectedness, larger protective effects were demonstrated for studies published in journals with lower impact factors (*z* = −0.070, *p* < .05).

3.6. Sample characteristics

Furthermore, for positive life outlook, larger protective effects were found in samples with fewer boys ($z = -0.038$, $p < .05$) and a higher proportion of youth from a minority background ($z = -0.017$, $p < .05$). For social connectedness, larger protective effects were demonstrated for samples with younger participants ($z = -0.064$, $p < .05$).

3.7. Outcome characteristics

Regarding the domain school achievement, larger protective effects were found for right-wing ($r = -0.232$) and religious ($r = -0.234$) ideology compared to other ideologies ($r = -0.148$, $p < .01$). Social connectedness factors were only protective for religious ideology ($r = -0.271$), not for right-wing or other ideologies ($r = -0.043$ and $r = -0.032$).

3.8. Subdomains

Social connectedness factors were protective for society/country factors (macro; $r = -0.187$), not for community/group factors (meso; $r = 0.034$, $p < .001$).

3.9. Sensitivity analyses

Sensitivity analyses were performed to assess each study's contribution to the overall effect, by rerunning the analyses 41 times with removing a different study each time (Duindam et al., 2020). Results showed that the overall effect remained significant after each analysis, demonstrating that none of the studies had an individual (disproportionate) impact on the overall result (Table 2). Furthermore, the overall effect was slightly higher when the two zero-assigned effect sizes were removed ($N = 192$ effect sizes, mean $r = -0.100$, $SE = 0.016$). Three extreme outliers were identified (highest: mean $z = 0.829$, lowest: mean $z = -0.758$ and -0.741). The three outliers were replaced with the next highest or lowest effect size that fell within the normal range (highest: mean $z = 0.332$, lowest: mean $z = -0.549$). Results demonstrated that the overall effect was slightly higher (mean $z = -0.099$, $SE = 0.015$). Overall, the sensitivity analyses did not reveal a (disproportionate) impact on the overall effect.

3.10. The file-drawer problem

The Rosenthal fail-safe test demonstrated a fail-safe N of 165,113, which exceeds the critical value (of 255; Rosenthal, 1979), suggesting that there is no file drawer problem or publication bias. The funnel plot (Fig. 2) demonstrates a slightly asymmetrical plot. However, the Egger's test revealed that there was no significant relationship between the standard error and effect size ($\beta = 0.053$, $SE = 0.42$, $p = .207$) and the trim and fill analysis estimated that zero effect sizes were missing, both on the left and right side of the distribution. Hence, this suggests that there is no publication bias.

4. Discussion

Important risk factors have been identified for extremism and violent radicalization, however, less is known about protective factors, even though they may be crucial for increasing the accuracy of risk assessment and optimizing prevention. A specific meta-analytic overview of protective factors in youth was not yet available, while youth are most susceptible for violent radicalization (Siegel et al., 2019). This meta-analysis builds on previous systematic work (Lösel et al., 2018) by testing the impact of protective factors for extremism and violent radicalization for the age group of 25-years or younger and conducting moderator analyses.

In total, 41 studies were included, yielding 49 independent samples

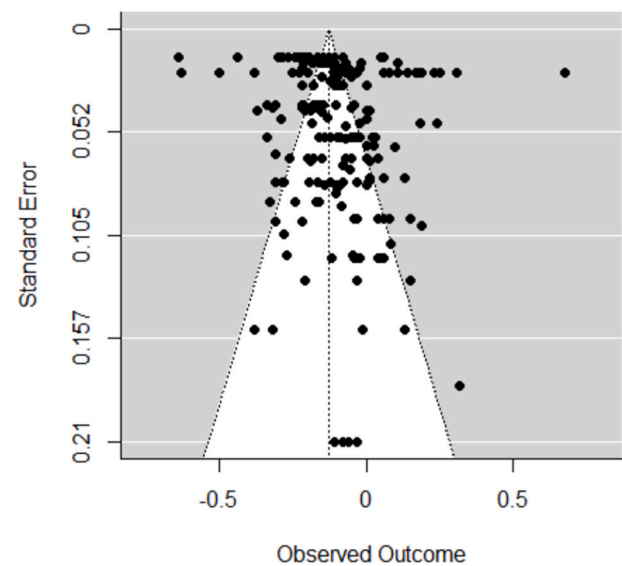


Fig. 2. Funnel plot.

and 194 effect sizes ($N = 55,563$ juveniles). Protective factors were classified into 12 protective domains. Overall, a small, negative impact of protective factors on (violent) radicalization ($r = -0.098$) was found.

4.1. Protective factor domain results

Six out of 12 protective domains yielded small to medium significant effects, that is (from largest to smallest) school achievement, female gender, (pro)social behavior, positive parental involvement, procedural justice, and positive intergroup contact. The protective effect found for the *school achievement* domain, which encompassed both academic success and a higher educational level, is consistent with previous research of protective factors against criminal behavior (e.g., Lösel & Farrington, 2012). However, the role that school achievement plays proves complex: there is contradicting evidence that suggests that individuals with a university degree are overrepresented among terrorists who have committed violent extremist acts (e.g., Gambetta & Hertog, 2019). It may be that school achievement usually protects by fostering stronger bonds to societal norms and society (i.e., Social Control Theory; Becker, 2021; Hirschi, 1969), but when opportunities are lacking for more highly educated individuals, relative deprivation may drive some towards violent radicalization. Another interpretation could be that particularly well-educated extremists are leaders with strong ideological bonds, but this relation was not analyzed in the primary studies.

Also, in line with research on delinquency and antisocial behavior, being *female* was protective against extremism and violent radicalization. It has been suggested that girls might be better protected against a delinquent path because they are ahead of boys in terms of their social-cognitive development (e.g., Bennett et al., 2005). Other explanations for the gender gap have proposed that girls might handle social stressors differently from boys (internalize versus externalize), girls might have better social bonds, and/or girls might have stronger self-control skills (Foster, 2014). The *(pro)social behavior* domain, reflecting a variety of characteristics that facilitate prosocial interaction (e.g., empathy, prosocial involvement, perspective taking), was also protective in line with protection against antisocial behavior more generally (Gubbels et al., 2023). A sought out target of anti-radicalization programs (e.g., Feddes et al., 2015), empathy and perspective taking, are deemed important buffers for (violent) radicalization because they have been related to less prejudice and stereotyping, which is important for successful intergroup contact (Pettigrew & Tropp, 2008). *Intergroup contact* itself, which encompassed non-extremist out-group friendships, multicultural acquisition, and other forms of positive identification between groups, was

also protective. Radical ideologies often include an outside group, which they hold responsible for negative outcomes (e.g., Pauwels & De Waele, 2014). Therefore, maintaining positive (non-deviant) out-group attitudes may act as a deterrent against being drawn into radicalization (Beelmann, 2020).

The protective effect of *positive parental involvement*, which consisted of parental control and bond features, is in line with prior research on the importance of parenting in various developmental outcomes (e.g., Jeong et al., 2021). No difference in protective impact was found for parental control versus bond features. Indeed it might be a warm parent-child relationship combined with parental control and limit-setting that is most effective in preventing violent radicalization (e.g., Sikkens et al., 2018). This finding is in accordance with the general research on positive effects of an authoritative parenting style (Baumrind, 2013). Finally, the protective domain of *procedural justice* was also protective against (violent) radicalization, which is in line with research on juvenile recidivism (e.g., Penner et al., 2014). Hence, an important intervention target may be to enhance juveniles' sense of trust in the justice system and acceptance of the law. Furthermore, justice officials might be trained to treat juveniles more fairly and respectful, or justice system procedures might be modified (e.g., enhance the quality of decision making, quality of treatment; Bottoms & Tankebe, 2012; Penner et al., 2014).

No significant effects were found for the six other protective domains, including religiosity, well-being, positive school environment, social connectedness, positive life outlook, and skills (e.g., self-control, conflict coping). It could be that it is the interplay of some of these factors that is important in radicalization (e.g., Soliman et al., 2016), not their independent effects. Similar to risk effects, protective effects seem to be stronger by aggregation according to a dose-response relationship (Lösel & Farrington, 2012). For example, a recent study in adolescents found that poor well-being alone was insufficient to catalyze violent radicalization. Rather it was the youth who felt disconnected from society due to their poor well-being that tended to disengage from authorities and gravitate towards violent radicalization (Costabile et al., 2020).

4.2. Moderating effects

Results of the moderator analyses showed that some study, sample, and outcome characteristics significantly moderated the effects. Even though *social connectedness* was only found to be a protective factor at trend-significance, moderator analyses demonstrated that macro-level factors, that is one's connection to society or country, were protective, whereas meso-level factors, related to the connectedness to one's group or community, were not. This is interesting as it is generally assumed that social systems most proximal to an individual have the biggest impact (e.g., Bronfenbrenner, 1979). Perhaps when it comes to (violent) radicalization though, the connection to society or the nation at large is a crucial protector, as radicalization involves fusion with a radical ingroup that try to isolate people from the social environment at large (Doosje et al., 2016).

For the domain of *positive life outlook*, factors were more protective in samples with fewer boys. Although still an underexplored area, drivers of radicalization can exhibit gender-specific patterns (e.g., Del Pino-Brunet et al., 2023). Initial research has found that certain factors such as mental illness, trauma, family dysfunction, and interpersonal rejection are more crucial as explanatory factors for female offending, compared to male offending (de Vogel & Nicholls, 2016). Consequently, if certain life adversities are more strongly related to (violent) radicalization in girls, cultivating positive feelings of contentment with life, both presently and in the future, along with fostering a positive identity, may become even more crucial for their protection.

Larger protective effects for *positive life outlook* were also found in samples with fewer youth from a minority background. A minority status has been associated with higher experiences of social adversity

and discrimination in society (Andriessen et al., 2019; Bussemakers & Kraaykamp, 2020); therefore, in a more challenging socioeconomic context *positive life outlook* may be less of a buffer for violent radicalization. This is in line with the more general criminological finding that a negative neighborhood context has an impact on late onset of offending even when individual protective factors are present (Wikström & Loeber, 2000).

Participants' age moderated the effect of the *social connectedness* protective domain, with larger effects for younger participants in this sample. Adolescence is an important phase for identity development, which takes place in the context of close relationships (Branje et al., 2021). Feeling socially connected has therefore been suggested as an important deterrent of negative outcomes for this age group (e.g., Culyba et al., 2019), in particular also because risk-taking behaviors are often influenced by pressure in the social environment (e.g., McCoy et al., 2019).

Furthermore, *school achievement* was more protective for right-wing and religious ideology compared to other, in this case left-wing, ideologies. Perhaps this is related to higher educational degrees being more common among left-wing extremists (Chermak & Gruenewald, 2015). *Social connectedness* was only protective for religious ideologies, compared to right-wing and other ideologies. Perhaps this is not surprising, given the nationalistic attitudes which are central to right-wing ideologies and the ties to one's country, which is included in this domain (Mudde, 2019). It should be noted that ideology did not play a moderating role for most outcomes, supporting previous assertions about similarities between developmental processes of radical groups with different ideologies (Ebner, 2017).

For the *positive life outlook* and *skills* domains, protective effects were only found in Western countries, not in non-Western ones. Conformingly, more evidence has become available over the last decades on how risk and protective factors are context-dependent (Ellis et al., 2021).

Finally, for some protective domains, characteristics related to the study's quality moderated the outcome. For *religiosity*, larger effects were found for older studies and larger effects were observed for *positive school environment* when study quality was lower. Similarly, the protective effect of *social connectedness* was larger for studies published in journals with lower impact factors. Overall, this suggests that the quality of the evidence for these protective domains is weaker (Luchini et al., 2021).

4.3. Implications

This meta-analysis identified several protective factor domains – and their impact – for (violent) radicalization, which helps increase our understanding of the etiology of extremism and violent radicalization by providing insight into the factors that have a direct protective effect. This meta-analysis also sheds light on the circumstances under which factors were most protective (e.g., social connectedness was more impactful for adolescents, compared to emerging adults). Overall, current findings can inform risk assessment for (violent) radicalization. Even in the presence of multiple risk factors, most youth do not become violently radicalized. Risk assessment instruments for violent radicalization have been critiqued for their overidentification of false positives (Sarma, 2017). Including more knowledge on protective factors might help improve the incremental predictive validity of risk assessment instruments (e.g., Robbé et al., 2013). Current study results indicate that factors such as school achievement, (pro)social behavior, and parental involvement might be important topics to include in risk assessment. Including these factors in policies to promote positive development in youth may also be helpful to avoid adverse outcomes.

In line with the Risk Need Responsivity model, prevention programs against violent radicalization can benefit from taking a multifactorial approach (Andrews & Bonta, 2017). Targeting dynamic protective factors with the biggest impact might strengthen preventative approaches. A review by Jugl, Lösel, et al. (2021) demonstrated that while existing

psychosocial prevention programs appeared promising against violent radicalization, only few controlled studies had been conducted on their effectiveness. These programs had had a wide variety of goals, ranging from reducing prejudices, and stimulating critical thinking, to reintegration in society. A more recent review and international survey showed a substantial increase in evaluations of psychosocial prevention programs against extremism and violent radicalization, although many implementations in practice are not yet based on well controlled empirical evidence (Jugl, Lösel, et al., 2023). Prevention efforts might benefit from integrating more primary research on protective factors. Based on the current findings, protective factors such as school achievement, procedural justice, and (pro)social behavior might be important targets to include in future prevention efforts. Targeting parental factors related to their involvement in youth's lives also seems important (e.g., Zych & Nasaescu, 2022). Although domains such as well-being (i.e., factors related to satisfaction with life) and skills (e.g., self-control, cognitive competence) were not found to be protective on average, it is important to disentangle these findings further because particular skills (e.g., self-control; Pauwels & Svensson, 2017) might be more important than others.

While existing prevention programs generally focused on a specific radical ideology (e.g., religious ideology; Jugl, Lösel, et al., 2021, 2023), moderator analyses demonstrated that for most protective factors no distinction was found based on ideology. This suggests that a similar prevention/intervention approach may be applied to different types of radical ideology, providing opportunities for more universal prevention programs. Programs might also benefit from taking a strength-based perspective more generally. For example, the Good Lives Model has advocated that it is essential to focus on people's strengths to reduce the risk of reoffending, enhance motivation and engagement, and develop more personalized intervention efforts (Mallion et al., 2020; Ward et al., 2007). A recent review also mentioned sports programs as a promising way to prevent violent extremism by fostering youth resilience and community belonging (Asif et al., 2025). However, as in other areas of crime prevention, there are only few evaluations of sports programs with sound designs (Jugl, Bender, et al., 2023).

Overall, most identified protective factors had a small impact on extremism and violent radicalization. Only school achievement had a medium effect. This is similar to a recent meta-analysis on protective factors for antisocial behavior in youth, which also found mostly small effects ($r < -0.20$) (Gubbels et al., 2023). While the majority of risk factors for (violent) radicalization are also small, some medium and larger ones have been identified (e.g., activism, perceived in-group superiority, Emmelkamp et al., 2020; Wolfowicz et al., 2021). It has been suggested that in many developmental domains the “bad is stronger than the good” (Baumeister et al., 2001), suggesting that the impact of risk factors might outweigh that of protective factors. However, what might be needed is a closer examination of the exact role protective factors play in interaction with risk factors, especially in terms of buffering the negative impact of risk factors (e.g., Lösel & Farrington, 2012). From a methodological perspective, it is difficult to explain substantially more outcome variance in a study once aggregated risks have already been included (Lösel & Bender, 2003).

4.4. Limitations and directions for future research

The current meta-analysis included 41 studies. Even after the extensive literature review that was employed, it cannot be assumed that all primary studies on protective factors were found. However, our search resulted in the inclusion of more studies than recent reviews on similar topics (e.g., Emmelkamp et al., 2020; Lösel et al., 2018). In addition, the Rosenthal fail-safe test and trim-and-fill analyses did not detect a publication bias. By employing a multilevel random effects model, we ensured all effect sizes could be included and no data were lost.

A limitation of this study, and meta-analytic research more

generally, is the merging of different type(s) of outcomes. This is especially challenging for research on violent radicalization and extremism, as many researchers employ different definitions for the constructs. For example, it is important to distinguish between violent versus non-violent outcomes, and also radicalized attitudes versus behavioral outcomes, as there are shared and unique protective factors (e.g., Wolfowicz et al., 2021). Especially for the prevention of violent extremist acts, it is important to have a clear understanding of protective factors for violently radical behavior specifically (as most people with radical attitudes will not become violent; Moskalenko & McCauley, 2021). Even though the current meta-analysis did look at violent versus non-violent outcomes, analyses could not be completed for all protective domains due to a lack of primary research. In addition, the impact of protective factors could not be examined for violent radical behaviors only, due to a lack of studies (behavioral outcomes were combined with intentions). Therefore, more research is needed on the role of protective factors that clearly define and include a variety of radicalization outcomes (i.e., attitudes, intentions, behaviors, violent, non-violent).

Furthermore, causality cannot be assumed as most studies were cross-sectional, not experimental or longitudinal. More research on protective factors against radicalization is needed, especially studies employing longitudinal and experimental designs across varied risk levels. It is also important that future primary studies more specifically examine the role that protective factors play in reducing risk for (violent) radicalization, beyond a direct reduction. In this regard it might be helpful for primary studies to employ research designs that can examine interrelations between multiple risk and protective factors, as it is the accumulation of risk factors – in the absence of protective factors – that is associated with violence (e.g., Stoddard et al., 2013). This type of knowledge can also test theory and more generally improve the understanding of the etiology of (violent) radicalization, which is essential for the development of prevention policies. Network analyses might be a helpful technique (e.g., Vial et al., 2020), which can help map the (strength of) relationships among risk and protective factors, give insight into their interaction, and identify which ones are most helpful/damaging for extremism and violent radicalization.

Finally, the protective factor domains tested in the current meta-analysis were limited. Almost all protective factors were from one ecological domain, focusing solely on the individual level. There are some particular areas that seem relevant to consider for a younger age group, and that have been found to play a role in extremism and (violent) radicalization before, such as the role of peers and media (e.g., Wolfowicz et al., 2022). Moreover, when looking at the identified protective factors for antisocial behavior more generally, future research should also consider including protective factors reflecting youth's attitudes towards self, others, and the world, as well as relationship quality (for an overview, see Gubbels et al., 2023).

5. Conclusion

This meta-analytical overview examined the conditional impact of various protective domain factors on extremism and violent radicalization in youth. School achievement, female gender, (pro)social behavior, positive parental involvement, procedural justice, and positive inter-group contact had medium to small protective effects. Findings suggest that risk assessment and prevention approaches can benefit from incorporating protective factors.

CRedit authorship contribution statement

Hanne M. Duindam: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Amanda W.G. van Loon:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Friedrich Lösel:** Writing – review & editing, Supervision, Methodology.

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Declaration of competing interest

None.

Appendices. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.avb.2025.102101>.

Data availability

Data will be made available on request.

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